

# Digital Logic ICs: 74HC Series, 595 Shift Register, CD4017, ULN2003, MCP3008, DS18B20

Document: EDU-DIGITAL-01 | Pinouts, Truth Tables & Application Circuits Reference

## 1. 74HC Series — CMOS Logic Gate ICs (Texas Instruments / NXP)

The 74HC series (High-Speed CMOS) is the foundational digital logic IC family. Supply Voltage: 2V to 6V (logic HIGH threshold =  $V_{CC} \times 0.7$ , logic LOW =  $V_{CC} \times 0.3$ ). Propagation Delay: 7ns at 5V. Fanout: Can drive 10 other 74HC inputs. Maximum current per output pin: 25mA. All inputs must be tied to VCC or GND; floating inputs cause unpredictable behavior and excessive current draw.

Common 74HC gate ICs: 74HC00 = Quad 2-input NAND gate (14-pin DIP). 74HC02 = Quad 2-input NOR gate. 74HC04 = Hex Inverter / NOT gate (14-pin DIP). 74HC08 = Quad 2-input AND gate. 74HC32 = Quad 2-input OR gate. 74HC86 = Quad 2-input XOR gate. 74HC74 = Dual D-type Flip-Flop with preset and clear.

**Gate Truth Tables:** **NAND(A,B):** 0+0=1, 0+1=1, 1+0=1, 1+1=0. **AND(A,B):** 0+0=0, 0+1=0, 1+0=0, 1+1=1. **NOR(A,B):** 0+0=1, 0+1=0, 1+0=0, 1+1=1. **XOR(A,B):** 0+0=0, 0+1=1, 1+0=1, 1+1=0. **Inverter(A):** 0=1, 1=0.

## 2. SN74HC595 — 8-bit Serial-In Parallel-Out Shift Register

Package: 16-pin DIP. One of the most useful ICs in Arduino projects: allows controlling 8 digital outputs using only 3 Arduino pins (SER, SRCLK, RCLK). Can be daisy-chained: connecting Q7S (serial output) of one 595 to SER of the next allows unlimited expansion with the same 3 pins. Output current: 35mA per pin.

Pinout: Pin 8=GND, Pin 16=VCC (5V). Pin 14=SER (serial data in), Pin 11=SRCLK (shift register clock), Pin 12=RCLK (storage/latch clock), Pin 10=SRCLR (shift register clear, active LOW, tie to VCC to disable). Pins 1-7,15=QA through QH (parallel outputs). Pin 9=Q7S (serial output for daisy-chaining).

**Arduino Code:** `#include <ShiftRegister.h> void setup() { pinMode(latchPin, OUTPUT); } void loop() { digitalWrite(latchPin, LOW); shiftOut(dataPin, clockPin, MSBFIRST, 0b10101010); digitalWrite(latchPin, HIGH); } This outputs alternating HIGH/LOW on the 8 output pins. The latch pin stores data into outputs only when pulsed HIGH.`

## 3. CD4017 — Johnson Decade Counter / Divider (Texas Instruments)

Package: 16-pin DIP. Supply: 3V to 15V (CMOS). Contains a 5-stage Johnson counter with 10 decoded outputs (Q0-Q9). Each clock pulse advances exactly one output HIGH while all others go LOW. Output sequence: Q0, Q1, Q2 ... Q9, then back to Q0. Used for LED chasers, sequential lighting, frequency division, and round-robin multiplexing.

Pinout: Pin 16=VCC, Pin 8=GND, Pin 14=CLK (clock input, rising edge triggered), Pin 13=CLK\_EN (Clock Enable, active LOW, tie to GND to enable), Pin 15=MR (Master Reset, active HIGH, tie to GND for normal operation), Pin 12=CO (Carry Out, goes HIGH for Q5-Q9, used for cascading), Pins 1-7,9-11=Q0-Q9 outputs (not in sequential pin order).

**Q Pin to Physical Pin mapping:** Q0=Pin3, Q1=Pin2, Q2=Pin4, Q3=Pin7, Q4=Pin10, Q5=Pin1, Q6=Pin5, Q7=Pin6, Q8=Pin9, Q9=Pin11. **LED Chaser:** Connect CLK to Arduino PWM or a 555 oscillator. Connect 10 LEDs with 220-ohm resistors from each Q output to GND. LEDs light sequentially with each clock pulse.

## 4. ULN2003A — Darlington Transistor Array (High-Current Driver)

Package: 16-pin DIP. Contains SEVEN NPN Darlington transistor pairs, each capable of switching 500mA at up to 50V with just 1.5V input. Each pair has a built-in suppression diode for inductive loads. Used to drive: stepper motors, relays, solenoids, lamps, and any load that exceeds Arduino GPIO current limits (>40mA).

Pinout: Pins 1-7 = IN1-IN7 (logic inputs, connect to Arduino GPIO pins). Pins 10-16 = OUT1-OUT7 (open-collector high-current outputs, connect to load). Pin 8 = GND. Pin 9 = COM (connect to positive supply voltage of the load being driven; the internal freewheeling diodes connect their anodes to COM).

28BYJ-48 Stepper Motor Driver: The ULN2003A is the standard driver for the 28BYJ-48 5V stepper motor. Connect IN1-IN4 to Arduino pins 8,9,10,11. Connect motor coil wires BLUE,PINK,YELLOW,ORANGE to OUT1-OUT4. Connect RED wire (5V) to COM. Use Stepper.h or AccelStepper library. 28BYJ-48 steps per revolution: 2048 (in half-step mode) or 4096 (in wave drive).

## 5. MCP3008 — 8-Channel 10-bit SPI ADC (Microchip Technology)

Package: 16-pin DIP. Converts 8 analog voltage inputs to digital values using SPI interface. Resolution: 10-bit (0-1023 values). Max Sample Rate: 200 kSPS at 5V. Supply: 2.7V to 5.5V. Reference Voltage: Selectable VDD or external VREF. This IC is essential for ESP32 and Raspberry Pi which have limited or no analog inputs.

Pinout: Pins 1-8 = CH0-CH7 (analog inputs). Pin 9 = DGND (digital ground), Pin 10 = CS/SHDN (chip select, active LOW), Pin 11 = DIN (SPI MOSI), Pin 12 = DOUT (SPI MISO), Pin 13 = CLK (SPI clock), Pin 14 = AGND (analog ground, connect to DGND), Pin 15 = VREF (reference voltage, connect to VCC for full-range), Pin 16 = VDD (2.7V-5.5V).

**SPI Read: Send 3 bytes: [0x01, (0x80 | channel<4), 0x00]. Response in bytes 2-3: result = ((byte2 & 0x03) << 8) | byte3. Voltage = result \* (VREF / 1023.0). Arduino: use SPI.transfer() or the MCP3XXX library.**

## 6. PCF8574 — 8-bit I2C GPIO Expander (NXP Semiconductors)

Package: 16-pin DIP or SOIC. Adds 8 quasi-bidirectional I/O pins accessible via I2C with just 2 Arduino wires (SDA + SCL). Supply: 2.5V to 6V. I2C Address: Configurable via 3 address pins (A0, A1, A2) giving 8 possible addresses (0x20-0x27). The PCF8574A variant uses addresses 0x38-0x3F (allowing both variants to coexist on the same bus, totalling 128 GPIOs).

This is the IC inside most I2C LCD backpack modules (the small board you solder to the 16x2 LCD to reduce wiring from 12 pins to 4). Also used in I2C keypad scanners and multi-relay boards. Each pin can source/sink 25mA. Writing 0xFF to the port register sets all pins HIGH (input mode). Writing 0x00 drives all LOW.

## 7. MAX232 — RS-232 to TTL Level Converter (Maxim / Texas Instruments)

Package: 16-pin DIP. Converts between RS-232 voltage levels (+/-3V to +/-15V) and TTL/CMOS logic levels (0V/5V). Contains two transmitter channels (TTL to RS-232) and two receiver channels (RS-232 to TTL). Supply: 5V. Internal charge pump generates +10V and -10V from the 5V supply using four 1uF external capacitors.

Required External Capacitors: C1 (1uF) between pins 1-3, C2 (1uF) between pins 4-5, C3 (1uF) between pin 6 and GND, C4 (1uF) between pin 2 and pin 16 (V+). TTL TX to RS-232: Pin 11 (T1IN) receives TTL, Pin 14 (T1OUT) outputs RS-232. RS-232 to TTL RX: Pin 13 (R1IN) receives RS-232, Pin 12 (R1OUT) outputs TTL. Essential for connecting older serial devices or programming AVR microcontrollers without USB-to-serial adapter.

## 8. DS18B20 — 1-Wire Digital Temperature Sensor (Maxim / Dallas)

Package: TO-92 (looks like a transistor, 3 pins). Resolution: 9 to 12-bit selectable (0.0625 degrees C at 12-bit). Temperature Range: -55 to +125 degrees Celsius. Accuracy: +/-0.5C from -10C to +85C. Communication: 1-Wire protocol (single data wire + GND + VCC). Each sensor has a unique 64-bit ROM address, allowing multiple sensors on the same single wire.

Pinout (TO-92, flat face towards you): Left pin = GND, Center pin = DQ (data line), Right pin = VDD (3V-5.5V). MANDATORY: 4.7k pull-up resistor between DQ and VDD. Without pull-up resistor, sensor communication fails completely. Parasitic power mode (powering through DQ line with only 2 wires) is possible but reduces reliability; always use 3-wire mode in student projects.

**Arduino Code: #include , #include . OneWire oneWire(pin); DallasTemperature sensors(&oneWire); sensors.begin(); sensors.requestTemperatures(); float tempC = sensors.getTempCByIndex(0); Serial.println(tempC);**